Related Work: Ontology-Grounded LLMs for Breast Cancer Misinformation Detection

Key Papers, Approaches, and Findings

A. Medical Fact-Checking via Structured Claim Verification

1. FAITH (Zhou et al., AAAI 2026, arXiv:2511.12817)\ *Approach:* Decomposes LLM responses into atomic claims, links each to a medical knowledge graph, scores via evidence paths. No reference answers needed.\ *Result:* KG-grounded evaluation correlates higher with clinician judgments than text-only; robust to paraphrases.

2. CER (Barone et al., SIGIR 2025, arXiv:2509.13888)\ *Approach:* Evidence retrieval + LLM reasoning + supervised veracity prediction. Grounds outputs in scientific evidence.\ *Result:* SOTA on HealthFC, BioASQ-7b, SciFact with cross-dataset generalization.

3. Atomic Fact-Checking for Medical QA (Vladika et al., 2025, arXiv:2505.24830)\ *Approach:* Decomposes responses into atomic facts, each independently verified against medical guidelines KB.\ *Result:* 40% answer improvement, 50% hallucination detection rate.

4. MedGraphRAG (Wu et al., 2024, arXiv:2408.04187)\ *Approach:* Triple-linked graph (documents → medical sources → controlled vocabularies) + U-Retrieval (top-down precise + bottom-up refinement).\ *Result:* Outperforms SOTA on 9 medical QA + 2 fact-checking benchmarks with source traceability.

5. TrumorGPT (Hang et al., 2025, arXiv:2505.07891)\ *Approach:* Semantic health KG from medical news + few-shot LLM for KG construction + GraphRAG for grounding.\ *Result:* Superior health fact-checking; distinguishes true rumors from false claims.

6. GraphCheck (Chen et al., 2025, arXiv:2502.16514)\ *Approach:* Extracts KGs from text, uses GNN as soft prompt for LLMs. Single-inference multi-hop reasoning.\ *Result:* 7.1% improvement over baselines; comparable to DeepSeek-V3/o1 with fewer parameters.

B. Ontology-Constrained Biomedical NLP

7. ANCHOR-RE (Ming et al., Aug 2026, arXiv:2608.03154)\ *Approach:* Neuro-symbolic framework: ontology-guided reasoning + external knowledge grounding + verification rules. Training-free.\ *Result:* F1 0.654→0.676 (SemRepGS), 0.769→0.872 (DDI). 69% precision on unseen 2026 literature.

8. RELATE (Olasunkanmi et al., 2025, arXiv:2509.19057)\ *Approach:* Three-stage: ontology predicate embeddings → SapBERT similarity retrieval → LLM reranking with explicit negation handling.\ *Result:* 94% accuracy@10; handles negated assertions; rejects 0.4% irrelevant associations.

9. ChronoMedKG (Ahmed et al., 2026, arXiv:2605.22734)\ *Approach:* Multi-agent LLM pipeline (multi-model consensus) builds temporal KG with 460K evidence-linked triples + credibility scoring.\ *Result:* KG retrieval rescues **47–65%** of LLM failures vs. 17–29% for standard RAG.

10. OncoNet Ontology (Karim et al., 2023, arXiv:2310.08365)\ *Approach:* Cancer-specific ontology (ONO) + BioBERT/SciBERT extractors + LLM fine-tuning for KG enrichment.\ *Focus:* Biomarker discovery and gene-disease validation (not misinformation detection).

C. LLM Behavior on Patient Misinformation

11. ThReadMed-QA (Munnangi & Savage, MLHC 2026, arXiv:2607.12884)\ *Approach:* 2,437 real patient-physician threads (8,204 QA pairs). Tests LLM misconception correction over multiple turns.\ *Result:* GPT-5/Claude correct misconceptions ~85% initially but **drop to ~50% within 2 follow-ups**. Error propagation dominates.

12. Medical Sycophancy (Ping et al., Aug 2026, arXiv:2608.01017)\ *Approach:* Factorial experiment: 4 factors x 5 models x 500 questions (1.2M trials). CoT trace analysis.\ *Result:* Fabricated sources raise sycophancy 2x; models reasoning about facts resist, those re-examining own answers concede.

13. Limited Impact of Medical Adaptation (Jeong et al., EMNLP 2024, arXiv:2411.08870)\ *Finding:* Domain-adapted medical LLMs perform WORSE than base models in 56.7% of clinical QA cases.\ *Implication:* Naive fine-tuning insufficient; external structured knowledge (ontology) is a better path.

D. Key Datasets

- **MedQuAD** (Ben Abacha & Demner-Fushman, BMC Bioinf. 2019): 47,457 QA pairs from 12 NIH sites
- **HealthFC** (Vladika et al., LREC-COLING 2024): 750 expert fact-checked health claims with evidence
- **FactKG** (Kim et al., ACL 2023): 108K claims with 5 reasoning types (one-hop, conjunction, multi-hop, negation)
- **CoVERT** (Mohr et al., LREC 2022): 300 COVID tweets with entities, relations, fact-check labels

E. Summary of Methodological Takeaways

1. **Claim decomposition works** — atomic verification yields 40–50% improvement (FAITH, Vladika)
2. **Ontology > general KG > RAG > LLM-only** — structured domain knowledge rescues 47–65% of failures

(ChronoMedKG)

3. **Negation modeling is essential** — medical misinfo often involves overstated/negated claims (RELATE)
4. **Patient language needs specific adaptation** — biomedical models alone insufficient (PVminer, ThReadMed-QA)
5. **LLMs degrade on multi-turn misconceptions** — 85%→50% accuracy; structured grounding prevents drift (ThReadMed-QA)
6. **Neuro-symbolic approach is training-free** — ontology constraints improve LLMs without fine-tuning (ANCHOR-RE)
7. **No prior work combines:** domain-specific cancer misinfo ontology + patient-language claim decomposition + structured verification